

Looped_Transformer

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1 Looped Transformer

1.1 理论

Looped Transformer 所有层共享相同的权重 θ ，每层的输入是前一层输出与初始输入的和：

$$\begin{aligned} h_0 &= \text{input} \\ h_l &= \text{TransformerBlock}(h_{l-1} + h_0 | \theta) \quad \text{for } l = 1, 2, \dots, L \end{aligned}$$

1.1.1 截断损失函数

$$\text{Loss} = \min_{\theta} \mathbb{E}_P \left[\frac{1}{b - b_0} \sum_{t=b_0}^b \frac{1}{k+1} \sum_{i=0}^k (Y_t(P^i | \theta) - f(x_{i+1}))^2 \right]$$

其中 - $\min_{\theta}$ ：这是我们在寻找一组最优的物理定律（模型参数 θ ）。 - $\mathbb{E}_P$ ：对所有生成的 Prompt 分布求期望。在统计物理中，这相当于求“系综平均”，即我们在无数种不同的初始条件（数据集）下测试这个定律是否普适。 - $\frac{1}{b-b_0} \sum_{t=b_0}^b$ ：这是时间平均。 b 是总迭代次数， T 是窗口大小， $b_0 = \max(b - T, 0)$ 。我们只关心系统演化到“稳态”附近的这段时间内（也就是 b_0 到 b 的表现），前面瞬态的误差我们不计较。 - $\frac{1}{k+1} \sum_{i=0}^k$ ：这是空间/序列平均。也就是在给定不同长度的上下文 P^i 时，模型预测下一个点的值与真实函数 $f(x_{i+1})$ 之间的均方误差。 - $Y_t(P^i | \theta)$ ：在时间 t 、给定历史信息 P^i 和系统参数 θ 时，系统当前的状态输出。

1.1.2 下游任务

- 数据生成方式：这篇论文解决的是纯粹的数学数据拟合问题。数据集是在训练循环中即时生成的浮点数向量。
- 以最基础的 **线性回归（Linear Regression）** 为例：
 - 维度设定：问题维度 $d = 20$ ，上下文样本数 $k = 40$ 。
 - 生成逻辑：每次迭代前，随机采样一个真实的参数 $w \sim \mathcal{N}(0, I_d)$ 。
 - 采样输入：生成 $k + 1$ 个输入样本（包含测试样本） $x_i \sim \mathcal{N}(0, I_d)$ 。

- 计算标签: $y_i = w^T x_i$ 。
- 拼接成 Prompt: 最终喂给模型的序列 P 是一个交替的序列 $(x_1, y_1, x_2, y_2, \dots, x_k, y_k, x_{test}, y_{dummy})$ 。
- 线性回归可以进一步改成非线性回归 (比如 ReLU 与 Linear 叠加), 或者更复杂的物理系统。

把 Prompt 映射到高维 Embedding 空间 双通道映射与“拉链式”交织

由于 x 和 y 代表不同的物理量 (一个是空间坐标/特征, 一个是目标观测值), 我们需要赋予它们独立的投影空间:

1. 定义两个独立的线性层:

```
read_in_x = nn.Linear(20, 256)
read_in_y = nn.Linear(1, 256)
```

2. 把前 k 个 x_i 丢给 `read_in_x`, 变成张量 X_{emb} , 形状为 $[batch, k, 256]$ 。
3. 把前 k 个 y_i 丢给 `read_in_y`, 变成张量 Y_{emb} , 形状为 $[batch, k, 256]$ 。
4. 像拉拉链一样, 在 `seq_len` 维度上把 X_{emb} 、 Y_{emb} 交织 (Interleave) 起来, 拼接成最终长度为 $2k$ 的提示向量 P 。

在工程实现中, 我们在 x_{test} 后填充了一个 y_{dummy} (如 0), 使实际拼接长度变为 $2k+2$ (偶数)。因此, 真正对 x_{test} 的预测结果位于输出序列的倒数第二个位置, 即索引 $[-2]$ 处。

1.2 代码

```
[163]: import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt
import time
```

1.2.1 绝对位置编码 (APE, Absolute Position Encoding)

```
[164]: class APE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096, b=10000):
        super().__init__()
        theta_i = 1 / (b ** (torch.arange(0, d_model, 2).float() / d_model)) # ↵
        ↪ [d_model/2]
        m = torch.arange(max_seq_len).float() # [max_seq_len]
        m_theta_i = torch.outer(m, theta_i) # [max_seq_len, d_model/2]
```

```

# APE 核心：直接生成一个完整的 pe 矩阵，偶数维度放 sin，奇数维度放 cos
stacked=torch.stack((torch.sin(m_theta_i), torch.cos(m_theta_i)),  

↳dim=-1) # [max_seq_len, d_model/2, 2]
pe=stacked.flatten(1,2) # [max_seq_len, d_model]
self.register_buffer('pe', pe.unsqueeze(0)) # shape: [1, max_seq_len,  

↳d_model]

def forward(self, x):
    # x shape: [batch_size, seq_len, d_model]
    return x + self.pe[:, :x.shape[1], :]

```

Learned APE

```

[165]: class LearnedAPE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096):
        super().__init__()
        self.pe = nn.Embedding(max_seq_len, d_model)

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        seq_len = x.shape[1]
        positions = torch.arange(seq_len, device=x.device)
        return x + self.pe(positions).unsqueeze(0)

```

自创：Multi-Scale Untied Positional Encoding (MS-UPE)

```

[166]: class MS_UPE(nn.Module):
    def __init__(self, num_heads, d_k, max_seq_len = 4096, b_0=10000,  

↳head_ratio=2):
        super().__init__()
        b=(b_0*(head_ratio**((torch.arange(0, num_heads))))).float() # [num_heads]
        # \theta_i = b^{-\frac{2i}{d}}, \quad i \in \{0, 1, \dots, \frac{d}{2}-1\}
        theta_i = 1/(b.unsqueeze(1)**(torch.arange(0,d_k,2).float()/d_k).  

↳unsqueeze(0)) # [num_heads, d_k/2]
        m = torch.arange(max_seq_len).float() # m = [0, 1, ..., seq_len-1]
        m_theta_i = torch.einsum('m,hk->hmk', m, theta_i) # [num_heads,  

↳seq_len, d_k/2]
        pe = torch.cat((torch.cos(m_theta_i), torch.sin(m_theta_i)), dim=-1) #  

↳[num_heads, seq_len, d_k]

```

```
self.register_buffer('pe', pe[None, :, :, :]) # 好处: pe 不需要更新参数,
注册为 buffer 后会自动放到正确的设备上
```

```
def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
    # x: [batch_size, num_heads, seq_len, d_k]
    seq_len = x.shape[2]
    pe = self.pe[:, :, :seq_len, :] # [1, num_heads, seq_len, d_k]
    x_out = x + pe
    return x_out
```

1.2.2 去掉 RoPE 的 Multi-Head Attention

注: APE 应该加在 ToyModel 上而不是 TransformerBlock 上, 因为我们只需要最开头输入处编码位置, 而不是叠加 num_blocks 次。

```
[167]: class MultiHeadAttention(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096):
        super().__init__()
        assert d_model % num_heads == 0, "d_model must be divisible by
        ↪ num_heads"
        # 维度属性
        self.num_heads = num_heads
        self.d_model = d_model
        self.d_k = d_model // num_heads
        # 权重矩阵
        self.W_Q = nn.Linear(d_model, d_model, bias=False)
        self.W_K = nn.Linear(d_model, d_model, bias=False)
        self.W_V = nn.Linear(d_model, d_model, bias=False)
        self.W_O = nn.Linear(d_model, d_model, bias=False)
        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
        ↪ tril (triangular lower) 提取下三角元素, 并把上三角元素置 0
        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,
        ↪ float('-inf')) # 下三角元素置 0, 上三角元素置 -inf
        self.register_buffer('mask', mask[None, None, :, :]) # [1, 1, seq_len,
        ↪ seq_len]
```

```

def forward(self, x): # x: [batch_size, seq_len, d_model]
    # 线性投影与分头
    batch_size, seq_len, _ = x.shape
    Q = self.W_Q(x).view(batch_size, seq_len, self.num_heads, self.d_k).
    ↪transpose(1, 2) # [batch_size, num_heads, seq_len, d_k]
    K = self.W_K(x).view(batch_size, seq_len, self.num_heads, self.d_k).
    ↪transpose(1, 2) # 交换 (1, 2) 是因为之后 softmax 时要进行矩阵乘法 (只乘后二维)
    V = self.W_V(x).view(batch_size, seq_len, self.num_heads, self.d_k).
    ↪transpose(1, 2) # 在 seq_len 维度上分配注意力
    # 缩放点积注意力与因果掩码 Causal Masking
    ## Query-Key 点积计算注意力得分
    scores = torch.matmul(Q, K.transpose(-2, -1)) / (self.d_k ** 0.5) # ↪
    ↪[batch_size, num_heads, seq_len, seq_len]
    ## 因果掩码和 softmax 计算注意力权重
    Attention = F.softmax(scores + self.mask[..., :seq_len, :seq_len], ↪
    ↪dim=-1) # [batch_size, num_heads, seq_len, seq_len]
    self.captured_attention = Attention # ** 新增 **: 捕获注意力矩阵, 用
hook 记录下来, 方便后续分析
    ## 乘以 Value 完成加权求和
    Out = torch.matmul(Attention, V) # [batch_size, num_heads, seq_len, ↪
    ↪d_k]
    # 多头拼接并乘以输出矩阵 (必须先内存连续化 (contiguous) 再 view, 否则会报错)
    Out = Out.transpose(1, 2).contiguous().view(batch_size, seq_len, -1) # ↪
    ↪[batch_size, seq_len, d_model]
    H = self.W_O(Out) # [batch_size, seq_len, d_model]
    return H

```

1.2.3 SwiGLU 和单个 Transformer Block 架构不变

```

[168]: class SwiGLU(nn.Module):
    def __init__(self, d_model, d_hidden):
        super().__init__()
        self.W = nn.Linear(d_model, d_hidden, bias=False)
        self.V = nn.Linear(d_model, d_hidden, bias=False)
        self.W2 = nn.Linear(d_hidden, d_model, bias=False)

```

```

def forward(self,x):
    x1 = F.silu(self.W(x))
    x2 = self.V(x)
    x_out = self.W2(x1 * x2)
    return x_out

```

```

[169]: class TransformerBlock(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096, multiplier=4,
    ↪norm_type='layernorm', ffn_type='gelu'):
        super().__init__()
        if norm_type == 'rmsnorm':
            self.norm1 = nn.RMSNorm(d_model) # 第一次归一化
            self.norm2 = nn.RMSNorm(d_model) # 第二次归一化
        elif norm_type == 'layernorm':
            self.norm1 = nn.LayerNorm(d_model) # 第一次归一化
            self.norm2 = nn.LayerNorm(d_model) # 第二次归一化
        self.attention = MultiHeadAttention(num_heads, d_model,
    ↪max_seq_len=max_seq_len)
        if ffn_type == 'swiglu':
            self.ffn = SwiGLU(d_model, d_model * multiplier)
        elif ffn_type == 'gelu':
            self.ffn = nn.Sequential(
                nn.Linear(d_model, d_model * multiplier),
                nn.GELU(),
                nn.Linear(d_model * multiplier, d_model)
            )

    def forward(self, x):
        # 第一次归一化 Pre-Norm
        x_norm1 = self.norm1(x)
        # 全局信息交互 Multi-Head Attention (MHA)
        h = self.attention(x_norm1)
        # 第一次残差连接
        x1 = x + h
        # 第二次归一化
        x_norm2 = self.norm2(x1)

```

```

# 前馈网络 Feed-Forward Network (FFN)
x_out = self.ffn(x_norm2)
# 第二次残差连接
x2 = x1 + x_out
return x2

```

1.2.4 保留 Attention Probe

```

[170]: class AttentionProbe:
    def __init__(self):
        self.captured_data = []
    def __call__(self, module, input, output):
        attention = module.captured_attention.detach().cpu().numpy()
        self.captured_data.append(attention)
    def reset(self):
        self.captured_data = []

```

1.2.5 Toy Model: Looped Transformer

```

[171]: class ToyModel(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, max_seq_len=4096,
                 norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
                 loop=True):
        super().__init__()
        self.loop = loop
        if loop:
            self.transformer_block = TransformerBlock(num_heads, d_model,
                ↪max_seq_len=max_seq_len, norm_type=norm_type, ffn_type=ffn_type) # 这里保证了
各层权重始终相同
            self.probe = AttentionProbe()
            self.transformer_block.attention.register_forward_hook(self.probe)
            self.captured_attention = None
        else:
            self.transformer_blocks = nn.ModuleList([
                TransformerBlock(num_heads, d_model, max_seq_len=max_seq_len,
                ↪norm_type=norm_type, ffn_type=ffn_type)
                for _ in range(num_blocks)
            ])

```

```

    ])
    self.probe = [AttentionProbe() for _ in range(num_blocks)]
    for block, probe in zip(self.transformer_blocks, self.probe):
        block.attention.register_forward_hook(probe)
    if pe_type == 'ape':
        self.ape = APE(d_model, max_seq_len=max_seq_len) # 位置编码只需要加在
最开始输入处
    elif pe_type == 'learned_ape':
        self.ape = LearnedAPE(d_model, max_seq_len=max_seq_len)
    self.num_blocks = num_blocks
    self.num_heads = num_heads
    self.d_model = d_model
    self.max_seq_len = max_seq_len

    def forward(self, x_0, num_eff=15, current_blocks=None):
        # num_eff 是需要加入误差计算的有效层数，也即理论公式中的  $T$ 
        # x 的形状为 [batch_size, seq_len, d_model]
        x_0 = self.ape(x_0) # 位置编码只加在最开始输入处
        x = x_0
        if current_blocks is None:
            current_blocks = self.num_blocks
        b_0 = max(0, current_blocks - num_eff) # 计算需要加入误差计算的有效层数对
应的起始层索引
        outputs = []
        if self.loop:
            self.probe.reset() # 每次前向传播前重置捕获的数据，避免混淆不同次前向
传播的注意力矩阵
            for i in range(current_blocks):
                if i == b_0:
                    x = x.detach() # 释放前面的计算图
                    x.requires_grad_(True)
                    x = self.transformer_block(x+x_0)
                    if i >= b_0:
                        outputs.append(x)
            self.captured_attention = list(self.probe.captured_data) # 显式 copy
一份数据，避免后续被修改

```

```

else:
    for probe in self.probe:
        probe.reset()
    for i, block in enumerate(self.transformer_blocks):
        if i == b_0:
            x = x.detach() # 释放前面的计算图
            x.requires_grad_(True)
            x = block(x+x_0)
            if i >= b_0:
                outputs.append(x)
        self.captured_attention = [probe.captured_data for probe in self.
→probe]
        outputs = torch.stack(outputs, dim=1) # [batch_size, num_eff, seq_len,
→d_model]
    return outputs

```

1.2.6 Downstream Task: Linear Regression

转换头

```

[172]: class RegressionHead(nn.Module):
    def __init__(self, d_model, d_x, d_y=1):
        super().__init__()
        self.read_in_x = nn.Linear(d_x, d_model)
        self.read_in_y = nn.Linear(d_y, d_model)
    def forward(self, x_data, y_data):
        # k 相当于 seq_len 的一半, 因为 x 和 y 交织在一起, 所以总的 seq_len 是 2k
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        x_emb = self.read_in_x(x_data) # [batch_size, k, d_model]
        y_emb = self.read_in_y(y_data) # [batch_size, k, d_model]
        # 交织成 (x_1, y_1, x_2, y_2, ..., x_k, y_k)
        stacked = torch.stack((x_emb, y_emb), dim=2) # [batch_size, k, 2,
→d_model]
        Prompt = stacked.flatten(1,2) # [batch_size, 2k, d_model]
        return Prompt

```

损失函数

```
[173]: class PredictionLoss(nn.Module):
    def __init__(self, d_model, d_y=1, loss_type='mse'):
        super().__init__()
        self.read_out = nn.Linear(d_model, d_y)
        if loss_type == 'mse':
            self.loss_fn = nn.MSELoss()
        elif loss_type == 'l1':
            self.loss_fn = nn.L1Loss()

    def forward(self, outputs, y_true, is_eval=False):
        # outputs: [batch_size, num_eff, seq_len, d_model]
        # y_true: [batch_size, seq_len/2, 1]
        if is_eval:
            y_outputs = outputs[:, -1, -2, :] # 只取最后一层的最后一个位置的输出
            # 作为预测值
            y_pred_final = self.read_out(y_outputs) # [batch_size, d_y]
            return y_pred_final
        y_outputs = outputs[:, :, 0::2, :] # [batch_size, num_eff, seq_len/2, d_model]
        y_preds = self.read_out(y_outputs) # [batch_size, num_eff, seq_len/2, d_y]
        loss = self.loss_fn(y_preds, y_true.unsqueeze(1).expand_as(y_preds)) # [batch_size, num_eff, seq_len/2, d_y]
        return loss
```

拼装

```
[174]: class RegressionSolver(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, d_x, d_y, max_seq_len,
                 norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
                 loop=True, loss_type='mse'):
        super().__init__()
        self.toy_model = ToyModel(num_blocks, num_heads, d_model, max_seq_len,
                                   norm_type=norm_type, ffn_type=ffn_type, pe_type=pe_type, loop=loop)
        self.head = RegressionHead(d_model, d_x, d_y)
        self.loss_fn = PredictionLoss(d_model, d_y, loss_type=loss_type)
```

```

    def forward(self, x_data, y_data, num_eff, current_blocks=None,
        ↪is_eval=False):
        Prompt = self.head(x_data, y_data) # [batch_size, seq_len, d_model]
        outputs = self.toy_model(Prompt, num_eff,
        ↪current_blocks=current_blocks) # [batch_size, num_eff, seq_len, d_model]
        output = self.loss_fn(outputs, y_data, is_eval=is_eval)
        return output

```

1.2.7 Experiment

1.2.8 数据生成

线性回归数据

```

[175]: def linear_data_generator(batch_size, seq_len, d_x, d_y, device='cpu'):
        w = torch.randn(batch_size, d_x, d_y, device=device) # 真实线性函数  $w$ 
        x_data = torch.randn(batch_size, seq_len//2, d_x, device=device) # 输入特
        征  $x$ 
        y_data = x_data @ w # 计算标签  $y = x @ w$ 
        return x_data, y_data

```

1.2.9 数据加载器

```

[176]: def dataloader(batch_size=64, seq_len=80, d_x=20, d_y=1, device='cpu',
        ↪data_type='linear'):
        while True:
            if data_type == 'linear':
                yield linear_data_generator(batch_size, seq_len, d_x, d_y,
                ↪device=device)

```

1.2.10 实验模块

```

[177]: class LoopedTransformerExperiment:
        def __init__(self, num_blocks, num_heads=8, d_model=256, lr=1e-4,
        ↪max_seq_len=4096, d_x=20, d_y=1, seed=None, experiment_name='default_exp',
            norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
        ↪optimizer_type='adam', loop=True, loss_type='mse'):
            # 设备选择: 优先使用 MPS (适用于 Apple Silicon), 其次是 CUDA (适用于
            ↪NVIDIA GPU), 最后退回 CPU
            if torch.backends.mps.is_available():

```

```

        self.device = torch.device("mps")
    elif torch.cuda.is_available():
        self.device = torch.device("cuda")
    else:
        self.device = torch.device("cpu")
    print(f"{experiment_name} initialized. Using device: {self.device}")
    if seed is not None:
        torch.manual_seed(seed)
    # 模型和优化器
    self.model = RegressionSolver(num_blocks, num_heads, d_model, d_x=d_x,
    ↪d_y=d_y, max_seq_len=max_seq_len,
                                norm_type=norm_type, ffn_type=ffn_type,
    ↪pe_type=pe_type, loop=loop, loss_type=loss_type).to(self.device)
    if optimizer_type == 'adam':
        self.optimizer = torch.optim.Adam(self.model.parameters(), lr=lr)
    elif optimizer_type == 'sgd':
        self.optimizer = torch.optim.SGD(self.model.parameters(), lr=lr)
    self.d_x = d_x
    self.d_y = d_y
    self.num_blocks = num_blocks

    def train(self, batch_size=64, seq_len=80, epochs=20, data_type='linear',
    ↪num_eff=15, scheduled_training=True, print_every=None, timing=False):
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        self.data_loader = dataloader(batch_size, seq_len, self.d_x, self.d_y,
    ↪device=self.device, data_type=data_type)
        self.loss_history = []
        num_eff = min(num_eff, self.num_blocks) # 确保 num_eff 不超过总层数
        if timing:
            start_time = time.time()
        for epoch in range(epochs):
            self.optimizer.zero_grad()
            x_data, y_data = next(self.data_loader) # 从数据加载器中获取一个批次
            的数据
            if scheduled_training:

```

```

        current_blocks = min(num_eff + epoch, self.num_blocks) # 随着训练的
        进行, 逐渐增加参与误差计算的有效层数

    else:
        current_blocks = self.num_blocks

        loss = self.model(x_data, y_data, num_eff=num_eff,
↪current_blocks=current_blocks, is_eval=False) # 前向传播计算损失 [batch_size,
↪num_eff, seq_len/2, d_y]

        loss = loss.mean() # 对所有维度求平均得到一个标量损失值
        loss.backward() # 反向传播计算梯度
        self.optimizer.step() # 更新模型参数
        self.loss_history.append(loss.item()) # 记录损失值
        if print_every is not None and (epoch + 1) % print_every == 0:
            print(f"Epoch {epoch+1}/{epochs}, Loss: {loss.item():.4f},
↪current_blocks: {current_blocks}")

        if timing:
            end_time = time.time()
            print(f"Training completed in {end_time - start_time:.2f} seconds.")

    def evaluate(self, batch_size=64, seq_len=80, data_type='linear',
↪loss_type='mse'):
        self.model.eval()
        with torch.no_grad():
            # 在评估模式下, 直接使用所有层的输出进行预测 (因为已经 no_grad 了, 所以
            不需要防止梯度爆炸了)

            x_eval, y_eval = next(dataloader(batch_size, seq_len+1, self.d_x,
↪self.d_y, device=self.device, data_type=data_type))

            y_pred = self.model(x_eval, y_eval, num_eff=self.num_blocks,
↪current_blocks=self.num_blocks, is_eval=True) # [batch_size, d_y]

            if loss_type == 'mse':
                mse=F.mse_loss(y_pred, y_eval[:, -1, :])
                print(f"MSELoss: {mse:.4f}") # 计算预测值与真实值之间的均方误差
            elif loss_type == 'l1':
                l1=F.l1_loss(y_pred, y_eval[:, -1, :])
                print(f"L1Loss: {l1:.4f}")

        return y_pred

```

1.2.11 主程序

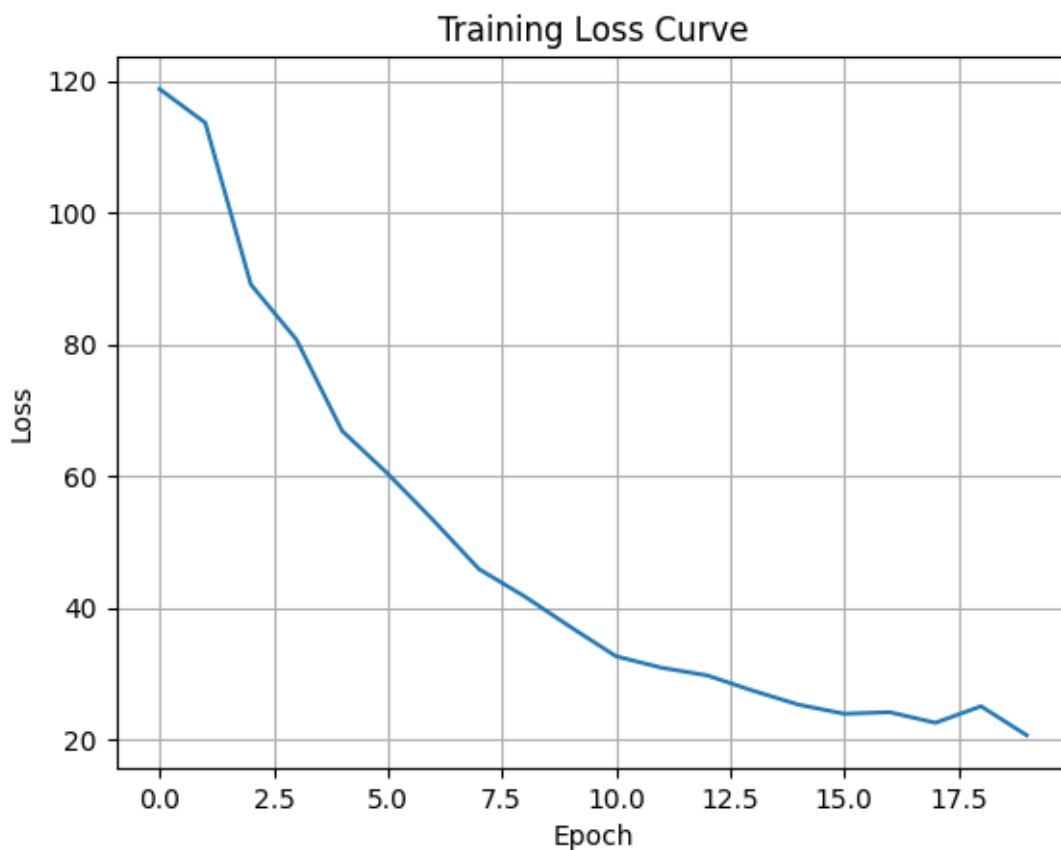
1.2.12 绘制 Loss 下降曲线

```
[178]: init_parameters = {
    'num_blocks': 20,
    'num_heads': 8,
    'd_model': 256,
    'lr': 1e-4,
    'max_seq_len': 4096,
    'd_x': 20,
    'd_y': 1,
    'seed': None,
    'experiment_name': 'Experiment',
    'norm_type': 'layernorm',
    'ffn_type': 'gelu',
    'pe_type': 'learned_ape',
    'loop': True,
    'loss_type': 'mse',
}

train_parameters = {
    'batch_size': 64,
    'seq_len': 80,
    'epochs': 20,
    'num_eff': 15,
    'data_type': 'linear',
    'scheduled_training': False,
    'print_every': 5,
    'timing': False
}

experiment = LoopedTransformerExperiment(**init_parameters)
experiment.train(**train_parameters)
plt.plot(experiment.loss_history)
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Training Loss Curve')
plt.grid()
plt.show()
```

Experiment initialized. Using device: mps
Epoch 5/20, Loss: 66.8336, current_blocks: 20
Epoch 10/20, Loss: 37.0737, current_blocks: 20
Epoch 15/20, Loss: 25.2541, current_blocks: 20
Epoch 20/20, Loss: 20.6124, current_blocks: 20



1.2.13 Scheduled Training vs Non-Scheduled Training 对比实验

```
[179]: init_parameters['seed'] = 42
init_parameters['experiment_name'] = 'Experiment scheduled'
train_parameters['scheduled_training'] = True
train_parameters['print_every'] = 2
train_parameters['timing'] = True
experiment1 = LoopedTransformerExperiment(**init_parameters)
experiment1.train(**train_parameters)
```

```

init_parameters['experiment_name'] = 'Experiment non-scheduled'
train_parameters['scheduled_training'] = False
experiment2 = LoopedTransformerExperiment(**init_parameters)
experiment2.train(**train_parameters)

# 绘制 Loss 下降曲线
import matplotlib.pyplot as plt
# 在一张图上绘制两条曲线，分别对应 scheduled_training=True 和 False 的情况
# 图例（不同颜色代表不同的训练方式）
plt.plot(experiment1.loss_history, label='Scheduled Training', color='blue')
plt.plot(experiment2.loss_history, label='Non-Scheduled Training', color='red')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Loss Curve - Scheduled Training vs Non-Scheduled Training')
plt.grid()
plt.legend()
plt.show()

```

Experiment scheduled initialized. Using device: mps

Epoch 2/20, Loss: 85.9484, current_blocks: 16

Epoch 4/20, Loss: 83.7361, current_blocks: 18

Epoch 6/20, Loss: 80.8645, current_blocks: 20

Epoch 8/20, Loss: 57.5536, current_blocks: 20

Epoch 10/20, Loss: 42.4516, current_blocks: 20

Epoch 12/20, Loss: 32.4875, current_blocks: 20

Epoch 14/20, Loss: 30.1276, current_blocks: 20

Epoch 16/20, Loss: 26.6545, current_blocks: 20

Epoch 18/20, Loss: 24.9089, current_blocks: 20

Epoch 20/20, Loss: 25.2995, current_blocks: 20

Training completed in 7.08 seconds.

Experiment non-scheduled initialized. Using device: mps

Epoch 2/20, Loss: 139.2216, current_blocks: 20

Epoch 4/20, Loss: 104.0805, current_blocks: 20

Epoch 6/20, Loss: 81.3494, current_blocks: 20

Epoch 8/20, Loss: 58.4295, current_blocks: 20

Epoch 10/20, Loss: 43.5169, current_blocks: 20

Epoch 12/20, Loss: 33.5656, current_blocks: 20
Epoch 14/20, Loss: 30.8815, current_blocks: 20
Epoch 16/20, Loss: 27.0990, current_blocks: 20
Epoch 18/20, Loss: 25.0254, current_blocks: 20
Epoch 20/20, Loss: 25.2887, current_blocks: 20
Training completed in 7.21 seconds.

